Exp 13: Demonstrate Network Forensics using Pcapxray Tool.

Aim:

To demonstrate network forensics by analyzing PCAP (Packet capture) files using the Pcapxray tool to visualize and investigate network traffic.

Procedure:

1) Clone Pcapxray: git clone https://github.com/srinivas11789/Pcapxray.git

2) Install dependencies: pip install dpkt tkinter requests

3) Click 'Analyze' to parse the packets.

4) Click individual nodes to show detailed IP and protocol info.

5) Review the protocol distribution chart.

6) Check the DNS query log and HTTP request list.

7) Generate the save the forensic report.

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# Installation
git clone https://github.com/Srinivas11789/PcapXray.git
cd PcapXray
pip3 install -r requirements.txt --break-system-packages

# Run PcapXray GUI
python3 Source/main.py

# Alternatively - analyze PCAP with scapy (command line)
from scapy.all import rdpcap, IP, TCP, UDP
pkts = rdpcap('capture.pcap')
for pkt in pkts[:20]:
    if IP in pkt:
        print(f'{pkt[IP].src} -> {pkt[IP].dst}  proto={pkt[IP].proto}')

# tcpdump-based analysis
tcpdump -r capture.pcap -n | head -20
tcpdump -r capture.pcap -n 'tcp port 80'
tcpdump -r capture.pcap -n 'dns'
```

Result:

Network forensics was successfully demonstrated using pcapxray. Network topology was visualized from PCAP data.